Theory of Value

An Axiomatic Analysis of Economic Equilibrium
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Chapter 1: Mathematics

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Continuous Functions

In this section, S denotes a subset of $\mathbb{R}^K$, T a subset of $\mathbb{R}^L$, (x^q) a sequence of points of S, and (y^q) a sequence of points of T.

Continuous Functions

Let f be a function from S to T, and consider a point $x^0 \in S$. The function f is continuous at the point x^0 if:

$$\text{“}x^q \to x^0, y^0 = f\left(x^0\right), y^q = f\left(x^q\right)\text{”} \text{ implies } \text{“}y^q \to y^0\text{”}$$

In other words. f is continuous at x^0 if the image of any sequence tending to x^0 is a sequence tending to the image of x^0.

Example: Let f be the real-valued function of a real variable (i.e., a function from $\mathbb{R}$ to $\mathbb{R}$) defined by $y = \frac{1}{x}$ if $x \neq 0$, and $y = 0$ if $x = 0$. The function f is continuous at every point of $\mathbb{R}$, with the exception of 0, as its graph readily suggests.

The function f is continuous on S if it is continuous at every point of S.

Continuous Functions

Let S_1, S_2, T be subsets of $\mathbb{R}^{K_1}$, $\mathbb{R}^{K_2}$, $\mathbb{R}^{L}$ respectively, f be a function from S_1 to S_2 and g be a function from S_2 to T. Define a function h from S_1 to T by $h(x) = g(f(x))$ for every x in S_1. It is immediate that:

(1) If f is continuous at the point x of S_1, and if g is continuous at the point $f(x)$ of S_2, then h is continuous at x.

Continuous Functions

For every $n = 1, 2, \cdots, N$, let T_n be a subset of $\mathbb{R}^{L_n}$ and consider the product $T = \prod_{n=1}^{N} T_n$. It is immediate that the projection on T_n is continuous on T.

Continuous Functions

With the notations of the last paragraph, let f_n be a function from S to T_n, and define a function f from S to T by $f(x) = (f_n(x))$ for every x in S, where $(f_n(x))$ denotes the N-tuple $(f_1(x), f_2(x), \cdots, f_N(x))$. It is immediate that:

(2) If every f_n is continuous at the point x of S, then f is continuous at x.

Continuous Functions

One can prove:

(3) A function f from S to T is continuous on S if and only if the inverse image of every set closed in T is closed in S.

Proposition: A mapping f from a metric space X to a metric space Y. The following statements are equivalent:

(1) f is continuous.

(2) For each open set G in Y, the set $f^{-1}(G)$ is an open set in X.

(3) For each closed set F in Y, the set $f^{-1}(F)$ is an closed set in X.

(4) For every set A in X, $f(\overline{A}) \subseteq \overline{f(A)}$.

(5) For every set B in Y, $\overline{f^{-1}(B)} \subseteq f^{-1}(\overline{B})$.

Continuous Functions

Proof: The function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$\|x - x_0\| < \delta \qquad \text{implies} \qquad \|f(x) - f(x_0)\| < \varepsilon,$$

or, the function f is continuous at the point x_0 if for any $\varepsilon > 0$, there exists $\delta > 0$ such that

$$x \in N_\delta(x_0) \qquad \text{implies} \qquad f(x) \in N_\varepsilon(f(x_0)),$$

that is, $f(N_\delta(x_0)) \subseteq N_\varepsilon(f(x_0))$, equivalently, the function f is continuous at the point x_0 if for any neighborhood V of $f(x_0)$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$.

Continuous Functions

Proof of (1) $\Longrightarrow$ (2): For each open set G in Y, we want to check that set $f^{-1}(G)$ is an open set in X.
For any $x_0 \in f^{-1}(G)$, $f(x_0) \in G$. For any neighborhood V of $f(x_0)$ with $f(x_0) \in V \subseteq G$, there exists neighborhood U of x_0 such that $f(U) \subseteq V$, then $x_0 \in U \subseteq f^{-1}(V) \subseteq f^{-1}(G)$.
Proof of (2) $\Longrightarrow$ (1): For any neighborhood V of $f(x_0)$, there exists neighborhood V' of $f(x_0)$ such that $f(x_0) \in V' \subseteq V$, then $x_0 \in f^{-1}(V') \subseteq f^{-1}(V)$ and $f^{-1}(V')$ is an open set.
Proof of (2) $\Longrightarrow$ (3): For each closed set F in Y, $G \equiv Y \setminus F$ is open, then
$f^{-1}(G) = f^{-1}(Y \setminus F) = f^{-1}(Y) \setminus f^{-1}(F) = X \setminus f^{-1}(F)$ is open, then the set $f^{-1}(F)$ is an closed set in X.
Proof of (3) $\Longrightarrow$ (2): For each open set G in Y, $F \equiv Y \setminus G$ is closed, then
$f^{-1}(F) = f^{-1}(Y \setminus G) = f^{-1}(Y) \setminus f^{-1}(G) = X \setminus f^{-1}(G)$ is closed, then the set $f^{-1}(G)$ is an open set in X.

Continuous Functions

Proof of (3) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq \overline{f(A)}$, then $A \subseteq f^{-1}\left(\overline{f(A)}\right)$ and $f^{-1}\left(\overline{f(A)}\right)$ is a closed set in X, hence $\overline{A} \subseteq f^{-1}\left(\overline{f(A)}\right)$. Thus $f\left(\overline{A}\right) \subseteq \overline{f(A)}$.

Proof of (4) $\Longrightarrow$ (3): For each closed set F in Y, define $A = f^{-1}(F)$, then $f(A) = f\left(f^{-1}(F)\right) \subseteq F$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)} \subseteq \overline{F} = F$. Thus $\overline{A} \subseteq f^{-1}(F) = A$, $A = \overline{A}$ is closed, that is, the set $f^{-1}(F)$ is an closed set in X.

Proof of (3) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq f^{-1}\left(\overline{B}\right)$ and $f^{-1}\left(\overline{B}\right)$ is a closed set in X, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (3): For each closed set F in Y, $\overline{f^{-1}(F)} \subseteq f^{-1}\left(\overline{F}\right) = f^{-1}(F)$, then the set $f^{-1}(F)$ is an closed set in X.

Proof of (4) $\Longrightarrow$ (5): For every set B in Y, $f^{-1}(B) \subseteq X$, then $f\left(\overline{f^{-1}(B)}\right) \subseteq \overline{f\left(f^{-1}(B)\right)} \subseteq \overline{B}$, hence $\overline{f^{-1}(B)} \subseteq f^{-1}\left(\overline{B}\right)$.

Proof of (5) $\Longrightarrow$ (4): For every set A in X, $f(A) \subseteq Y$, then $\overline{A} \subseteq \overline{f^{-1}(f(A))} \subseteq f^{-1}\left(\overline{f(A)}\right)$, hence $f\left(\overline{A}\right) \subseteq \overline{f(A)}$. $\square$

Continuous Functions

In the important particular case where f is a real-valued function on S. i.e., where f is a function from S to $\mathbb{R}$, one can prove:

(3′) A function from S to $\mathbb{R}$ is continuous on S if and only if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S.

Example: (1) Let $\mathbb{R}_+$ be the set of positive reals, and consider the function f from $S = \mathbb{R} \times \mathbb{R}_+$ to $\mathbb{R}$ defined by $y = f(u, v) = \frac{u}{v}$. "The function f is continuous on S" is equivalent to "the inverse image of $(-\infty, y]$, i.e, the set $\{(u, v) \in S \mid u - yv \leqq 0\}$, is closed in S, and similarly for $[y, \infty)$; the second assertion is readily suggested by a drawing in $\mathbb{R}^2$, and easily proved. (2) The example of 1.7.b does not satisfy the continuity criterion of (3′), as the graph shows.

Continuous Functions

Proof: If A function from S to $\mathbb{R}$ is continuous on S, as every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in $\mathbb{R}$, then the inverse image is also closed in S.

Conversely, if the inverse image of every interval of $\mathbb{R}$ of the form $(-\infty, y]$ or $[y, \infty)$ is closed in S. Assume the function is $f : S \to \mathbb{R}$. So for any open interval of $\mathbb{R}$, the form (a, b), (a, ∞), $(-\infty, a)$, we have

$$f^{-1}(-\infty, a) = f^{-1}(\mathbb{R}) \setminus f^{-1}[a, \infty)$$
$$f^{-1}(a, \infty) = f^{-1}(\mathbb{R}) \setminus f^{-1}(-\infty, a]$$
$$f^{-1}(a, b) = f^{-1}(-\infty, b) \setminus f^{-1}(-\infty, a]$$

are all open in S.

We also have following lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals. So for any open set $O \subseteq \mathbb{R}$, there exists countable open interval $\{I_n; n \in \mathbb{N}\}$ such that $O = \bigcup_{n=1}^{\infty} I_n$. So

$$f^{-1}(O) = \bigcup_{n=1}^{\infty} f^{-1}(I_n)$$

are also open in S. Hence f is continuous. $\square$

Continuous Functions

Lemma: Every open set in $\mathbb{R}$ is the countable union of disjoint open intervals.

Proof: Let G be a non-empty open set in $\mathbb{R}$. Write $a \sim b$ if the closed interval $[a, b]$ if $a < b$ or $[b, a]$ if $b < a$, lies in G. This is an equivalence relation, in particular $a \sim b$ since $\{a\}$ is itself a closed interval. G is therefore the union of disjoint equivalence classes. Let $C(a)$ be the equivalence class containing a. Let $G = \bigcup_{a \in I} C(a)$ where $C(a) \neq C(a')$ if $a \neq a'$. $C(a)$ is clearly an interval. Also $C(a)$ is open, for if $k \in C(a)$, then $(k - \varepsilon, k + \varepsilon) \subseteq G$ for sufficiently small ε. But then $(k - \varepsilon, k + \varepsilon) \subseteq C(a)$; so G is the union of disjoint intervals. For each $C(a)$ choose a rational q_a, this map $I \to \mathbb{Q} : G(a) \longmapsto q_a$ is injective, so I is countable. $\square$

Continuous Functions

One can prove:

(4) Let f be a function from S to T. If f is continuous on S, and if S is compact, then $f(S)$ is compact.

Proof: Let $\bigcup_{\gamma\in\Gamma} U_\gamma$ be an open covering of $f(S)$ on T. Then the sets $f^{-1}(U_\gamma)$ are open and $\left\{f^{-1}(U_\gamma); \gamma\in\Gamma\right\}$ form an open covering of S since we have show

$$f^{-1}\left(\bigcup_{\gamma\in\Gamma} U_\gamma\right) = \bigcup_{\gamma\in\Gamma} f^{-1}(U_\gamma)$$

Since S is compact, there exists a finite subset $\Gamma'\subseteq\Gamma$ such that $\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)$ still forms a covering of S, then

$$f(S) = f\left(\bigcup_{\gamma\in\Gamma'} f^{-1}(U_\gamma)\right) = f\left(f^{-1}\left(\bigcup_{\gamma\in\Gamma'} U_\gamma\right)\right) \subseteq \bigcup_{\gamma\in\Gamma'} U_\gamma$$

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately:

(4′) (Weierstrass' Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is non-empty compact, then $f(S)$ has a maximum and a minimum.

Proof: $f(S)$ is a compact subset of $\mathbb{R}$, hence closed and bounded. Now, any bounded set $A \subseteq \mathbb{R}$ has a least upper bound $\sup A$ and a greatest lower bound $\inf A$, and these two points belong to the closure A. But applying this to $A = f(S)$, which is closed, we conclude that $\sup f(S)$ and $\inf f(S)$ belong to $f(S)$ itself, which is exactly what is being claimed. $\square$

Continuous Functions

"$f(S)$ has a maximum" also means (1.5.g) "there is an x^M in S such that for all $x \in S$ one has $f(x) \leqq f\left(x^M\right)$." Such an element is called a maximizer of f. "$f(S)$ has a minimum" also means (1.5.g) "there is an x^m in S such that for all $x \in S$ one has $f\left(x^m\right) \leqq f(x)$." Such an element is called a minimizer of f. The maximum (resp. minimum) of the set $f(S)$ is also called the maximum (resp. minimum) of the function f on S.

Example: Let f be the function defined in the example of 1.7.b, and I an interval to which the variable x in restricted. One sees on the graph of f how, when I is not bounded, or not closed, or owns 0 and a different point, f has no maximizer or no minimizer.

Continuous Functions

It is easy to prove (using (3)),

(5) Let f be a function from S to T. If f is continuous on S, and if S is connected, then $f(S)$ is connected.

Proof: If $f(S)$ is not connected in Y, then we can find two non-empty and disjoint open subsets V_1 and V_2 of $f(X)$ such that $V_1 \cup V_2 = f(S)$. But then, $f^{-1}(V_1)$ and $f^{-1}(V_2)$ are nonempty open subsets of S. Moreover, these sets are disjoint, and we have $f^{-1}(V_1) \cup f^{-1}(V_2) = f^{-1}(V_1 \cup V_2) = S$. Thus S is not connected. Contradiction! $\square$

Continuous Functions

Applying this result to the particular case where f is real-valued, one obtains immediately (see (8) of 1.6.u):

(5′) (Bolzano's Theorem) Let f be a function from S to $\mathbb{R}$. If f is continuous on S, and if S is connected, then $f(S)$ is an interval.

Thus, if x^1, x^2 are two points of S, and y is a real number such that $f\left(x^1\right) \leqq y \leqq f\left(x^2\right)$, then there is an $x \in S$ such that $f(x) = y$. In other words, f takes on all values between $f\left(x^1\right)$ and $f\left(x^2\right)$. This result is often applied to the case where S is an interval of $\mathbb{R}$.

The Intermediate Value Theorem. Let X be a connected metric space and $f : X \to \mathbb{R}$ a continuous function. If $f(x_1) \leqslant \alpha \leqslant f(x_2)$ for some $x_1 \in X$ and $x_2 \in X$, then there exists a $x \in X$ such that $f(x) = \alpha$.

Proof. $f(X)$ is connected in $\mathbb{R}$. Hence $f(X)$ must be an interval. □

Notions of Continuity for Functions

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f : X \to Y$ be a function. When it is clear to which space a pair of points belongs, I may drop the subscripts on d_X and d_Y and simply write d for the metric. I will also use the notion $B_\varepsilon(x)$ to denote the open ball of radius ε centered at x when the metric is obvious.

Recall that the Euclidean metric on $\mathbb{R}^N$ (or l^2) is given by

$$d_2\left(x^1, x^2\right) = \sqrt{\sum_{n=1}^{N} \left(x_n^1 - x_n^2\right)^2}$$

Other useful metrics on $\mathbb{R}^N$ include the max (or l^∞) metric

$$d_\infty\left(x^1, x^2\right) = \max\left\{\left|x_n^1 - x_n^2\right| : 1 \leqslant n \leqslant N\right\}$$

and the taxicab (or l^1) metric

$$d_1\left(x^1, x^2\right) = \sum_{n=1}^{N} \left|x_n^1 - x_n^2\right|$$

Notions of Continuity for Functions

Definition 1 (Continuity using neighborhoods). A function $f : X \to Y$ is continuous at x if for every neighborhood V of $f(x)$, its inverse image $f^{-1}(V)$ is a neighborhood of x. That is, for every open set V with $f(x) \in V$, there exists an open set U with $x \in U$ such that

$$z \in U \Longrightarrow f(z) \in V,$$

i.e., $f(U) \subseteq V$.

Note that this definition is topological, that is, it uses only the notion of neighborhoods and does not mention the metrics. The next definition makes explicit use of the metrics.

Notions of Continuity for Functions

Definition 2 (Continuity using metrics). A function $f : X \to Y$ is continuous at x if for every $\varepsilon > 0$ there exists some $\delta > 0$ such that

$$d_X(x, z) < \delta \Longrightarrow d_Y(f(x), f(z)) < \varepsilon.$$

Definition 3 (Continuity using sequences). A function $f : X \to Y$ is continuous at x if whenever $x_n \to x$, then $f(x_n) \to f(x)$.

Proposition 1. A function f is continuous at x in the topological sense of Definition 1 if and only if it is continuous at x in the metric sense of Definition 2 if and only if it is continuous at x in the sequential sense of Definition 3.

Numerical Functions Defined on a Topological Space

The ideas of continuity introduced above apply in particular to numerical functions and we say that a numerical function f, defined on a topological space X, is continuous at a point x_0 if it is a mapping of X into $\mathbb{R}$ and, as such, is continuous at x_0; in other words if, to each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad |f(x) - f(x_0)| < \varepsilon.$$

Numerical Functions Defined on a Topological Space

In certain situations this idea turns out to be too restrictive and we are forced to consider the concepts of lower and upper semi-continuity, as given in the following definitions.

A numerical function f defined on X is said to be lower semi-continuous (l.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) > f(x_0) - \varepsilon.$$

A numerical function f defined on X is said to be upper semi-continuous (u.s.c.) at a point x_0 if, for each $\varepsilon > 0$, there corresponds a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f(x) < f(x_0) + \varepsilon.$$

We note that f is continuous at x_0 if and only if it is both lower and upper semi-continuous at that point.

Numerical Functions Defined on a Topological Space

Properties of upper semi-continuous functions can easily be deduced from those of lower semi-continuous functions by replacing $f(x)$ by $-f(x)$; we shall therefore consider only the case of lower semi-continuous functions.

We say that a function f is lower semi-continuous (l.s.c.) (in X) if it is lower semi-continuous at each point of X.

We say that a function f is upper semi-continuous (u.s.c.) (in X) if it is upper semi-continuous at each point of X.

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open.

Proof. The family $\mathscr{D}^-$ of intervals

$$\{(\lambda, +\infty) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of lower semi-continuity that f is lower semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^-)$. Since $S_\lambda^- = f^{-1}((\lambda, +\infty))$, the theorem stated is equivalent to this result. $\square$

Corollary 1. A numerical function f is lower semi-continuous if and only if, for each number λ, the lower contour set $T_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open.

Proof. The family $\mathscr{D}^+$ of intervals

$$\{(-\infty, \lambda) \mid \lambda \in \mathbb{R}\}$$

determines a topology in $\mathbb{R}$. Then it follows quickly from the definition of upper semi-continuity that f is upper semi-continuous if and only if it is a continuous mapping of X into the topological space $(\mathbb{R}, \mathscr{D}^+)$. Since $S_\lambda^+ = f^{-1}((-\infty, \lambda))$, the theorem stated is equivalent to this result. $\square$

Corollary 1. A numerical function f is upper semi-continuous if and only if, for each number λ, the upper contour set $T_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be lower semi-continuous is that, for each number λ, the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is open, i.e. the lower contour set $T_\lambda^- = \mathbf{C}_X S_\lambda^- = \{x \mid f(x) \leqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)-\epsilon}^- = \{x \in X \mid f(x) > f(x_0) - \epsilon\}$ is an open set and $x \in S_{f(x_0)-\epsilon}^-$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)-\epsilon}^-$, then $f(x) > f(x_0) - \epsilon$ for all $x \in U$. Hence f is lower semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is lower semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^- = \{x \in X \mid f(x) > \lambda\}$. For any $x_0 \in S_\lambda^-$, $f(x_0) > \lambda$, then let $\epsilon = \dfrac{f(x_0) - \lambda}{2}$, there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $f(x_0) - \epsilon = \dfrac{f(x_0) + \lambda}{2} > \lambda$, $U \subseteq S_\lambda^-$, then S_λ^- is open. $\square$

Numerical Functions Defined on a Topological Space

Theorem 1. A necessary and sufficient condition for a numerical function f to be upper semi-continuous is that, for each number λ, the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is open, i.e. the upper contour set $T_\lambda^+ = \mathbf{C}_X S_\lambda^+ = \{x \mid f(x) \geqq \lambda\}$ is closed.

Proof. If the set $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$ is an open set for every $\lambda \in \mathbb{R}$, consider any $x_0 \in X$, then for any $\epsilon > 0$, $S_{f(x_0)+\epsilon}^+ = \{x \in X \mid f(x) < f(x_0) + \epsilon\}$ is an open set and $x \in S_{f(x_0)+\epsilon}^+$, then there exists a neighborhood U of x_0 such that $U \subseteq S_{f(x_0)+\epsilon}^+$, then $f(x) < f(x_0) + \epsilon$ for all $x \in U$. Hence f is upper semi-continuous.

Conversely, if $f : X \to \mathbb{R}$ is upper semi-continuous, then for any $\lambda \in \mathbb{R}$, let $S_\lambda^+ = \{x \in X \mid f(x) < \lambda\}$. For any $x_0 \in S_\lambda^+$, $f(x_0) < \lambda$, then let $\epsilon = \dfrac{\lambda - f(x_0)}{2}$, there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $f(x_0) + \epsilon = \dfrac{f(x_0) + \lambda}{2} < \lambda$, $U \subseteq S_\lambda^+$, then S_λ^+ is open. $\square$

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a lower semi-continuous function f attains in K the value $m = \inf_{x \in K} f(x)$ (in other words, $\inf_{x \in K} f(x)$ is a minimum).

Proof. Let μ be a number such that $\mu > m$. Then $S_\mu^- = \{x \in K \mid f(x) \leqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu > m} S_\mu^- \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = m$. $\square$

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a lower semi-continuous function on a compact metric space X. Then f is bounded below and attains its minimum.

Proof. Let $U_\lambda^- = (\lambda, +\infty)$. $\left\{U_\lambda^-; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^-\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^-\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^-\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup\limits_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^-\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^-\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^-$$

hence $f(x) \geqq \min\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded below.

Numerical Functions Defined on a Topological Space

Let $m = \inf_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is lower semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x_0) - \epsilon < f(x)$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_0) - \epsilon < f(x_{n_k})$$

let $k \to \infty$, we have

$$f(x_0) - \epsilon \leqq m.$$

As ϵ is arbitrary, we have $f(x_0) \leqq m$, and by definition of m, we have $f(x_0) \geqq m$, hence x_0 is a minimizer. $\square$

Numerical Functions Defined on a Topological Space

Theorem 2. If K is a compact subset of X, a upper semi-continuous function f attains in K the value $M = \sup_{x \in K} f(x)$ (in other words, $\sup_{x \in K} f(x)$ is a maximum).

Proof. Let μ be a number such that $\mu < M$. Then $S_\mu^+ = \{x \in K \mid f(x) \geqq \mu\}$ is not empty and is a closed subset of the compact set K; hence, by the finite intersection axiom, we have

$$\bigcap_{\mu < M} S_\mu^+ \neq \varnothing.$$

For any point x_0 of this intersection, we have $f(x_0) = M$. $\square$

Numerical Functions Defined on a Topological Space

Theorem 2. Let f be a upper semi-continuous function on a compact metric space X. Then f is bounded above and attains its maximum.

Proof. Let $U_\lambda^+ = (-\infty, \lambda)$. $\left\{U_\lambda^+; \lambda \in \Lambda\right\}$ be an open covering of $f(X)$. Then the sets $\left\{f^{-1}\left(U_\lambda^+\right); \lambda \in \Lambda\right\}$ form an open covering of X since we have show

$$f^{-1}\left(\bigcup_{\lambda\in\Lambda} U_\lambda^+\right) = \bigcup_{\lambda\in\Lambda} f^{-1}\left(U_\lambda^+\right).$$

Since X is compact, there exists a finite subset $\Lambda' = \{\lambda_1, \cdots, \lambda_N\}$ such that $\bigcup\limits_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)$ still forms a covering of X, then we have

$$f(X) = f\left(\bigcup_{\lambda\in\Lambda'} f^{-1}\left(U_\lambda^+\right)\right) = f\left(f^{-1}\left(\bigcup_{\lambda\in\Lambda'} U_\lambda^+\right)\right) \subseteq \bigcup_{\lambda\in\Lambda'} U_\lambda^+$$

hence $f(x) \leqq \max\{\lambda_1, \cdots, \lambda_N\}$ for $x \in X$. Then f is bounded above.

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Let $M = \sup_{x \in X} f(x)$, there exists a sequence $x_n \in X$ such that $f(x_n) \to m$. Since $x_n \in X$ and X is compact, we have some $x_0 \in X$ and some subsequence such that $x_{n_k} \to x_0$ as X is compact, and hence sequentially compact. As f is upper semi-continuous, for every $\epsilon > 0$ there exists a neighborhood U of x_0 such that $f(x) < f(x_0) + \epsilon$ for all $x \in U$, as $x_{n_k} \to x_0$, there exists $K \in \mathbb{N}$ such that for all $k \geqq K$, $x_{n_k} \in U$, hence

$$f(x_{n_k}) < f(x_0) + \epsilon$$

let $k \to \infty$, we have

$$M \leqq f(x_0) + \epsilon.$$

As ϵ is arbitrary, we have $M \leqq f(x_0)$, and by definition of M, we have $f(x_0) \leqq M$, hence x_0 is a maximizer. $\square$

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of lower semi-continuous functions, the function g such that $g(x) = \sup_{i \in I} f_i(x)$ is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) > g(x_0) - \varepsilon.$$

Since f_i is lower semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f_i(x) > f_i(x_0) - \varepsilon > g(x_0) - 2\varepsilon$$

and therefore

$$x \in U(x_0) \quad \Longrightarrow \quad g(x) \geqq g(x_0) - 2\varepsilon.$$

Hence g is lower semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. $\square$

Numerical Functions Defined on a Topological Space

Theorem 3. If $\{f_i \mid i \in I\}$ is a family of upper semi-continuous functions, the function g such that $g(x) = \inf_{i \in I} f_i(x)$ is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists an index i such that

$$f_i(x_0) < g(x_0) + \varepsilon.$$

Since f_i is upper semi-continuous, there exists a neighbourhood $U(x_0)$ such that

$$x \in U(x_0) \quad \Longrightarrow \quad f_i(x) < f_i(x_0) + \varepsilon < g(x_0) + 2\varepsilon$$

and therefore

$$x \in U(x_0) \quad \Longrightarrow \quad g(x) \leqq g(x_0) + 2\varepsilon.$$

Hence g is upper semi-continuous at x_0; since x_0 is arbitrary, the theorem is proved. $\square$

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are lower semi-continuous functions then the function $g(x) = \inf_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is lower semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) > f_n(x_0) - \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) > f_n(x_0) - \varepsilon,$$

whence

$$g(x) = \inf_{n \in I} f_n(x) \geqq \inf_{n \in I} \{f_n(x_0) - \varepsilon\} = g(x_0) - \varepsilon$$

and so g is lower semi-continuous. $\square$

Numerical Functions Defined on a Topological Space

Theorem 4. If $f_1, \cdots, f_N$ are upper semi-continuous functions then the function $g(x) = \sup_{n \in I} f_n(x)$ (where $I = \{1, \cdots, N\}$) is upper semi-continuous.

Proof. Let x_0 be any point of X. Given $\varepsilon > 0$, there exists a neighbourhood $U_n(x_0)$ such that

$$x \in U_n(x_0) \quad \Longrightarrow \quad f_n(x) < f_n(x_0) + \varepsilon.$$

If $x \in \bigcap_{n \in I} U_n(x_0)$, then, for all n,

$$f_n(x) < f_n(x_0) + \varepsilon,$$

whence

$$g(x) = \sup_{n \in I} f_n(x) \leqq \sup_{n \in I} \{f_n(x_0) + \varepsilon\} = g(x_0) + \varepsilon$$

and so g is upper semi-continuous. $\square$

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two lower semi-continuous numerical functions, the function $f_1 + f_2$ is lower semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) > a\} = \bigcup_{\lambda} \{x \mid f_1(x) > a - \lambda\} \cap \{x \mid f_2(x) > \lambda\}$$

is a union of open sets and so is open. $\square$

Numerical Functions Defined on a Topological Space

Theorem 5. If f_1 and f_2 are two upper semi-continuous numerical functions, the function $f_1 + f_2$ is upper semi-continuous.

Proof. The set

$$\{x \mid f_1(x) + f_2(x) < a\} = \bigcup_{\lambda} \{x \mid f_1(x) < a - \lambda\} \cap \{x \mid f_2(x) < \lambda\}$$

is a union of open sets and so is open. $\square$

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive lower semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is lower semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) > \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) > \lambda\} \cap \left\{x \mid f_2(x) > \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. $\square$

Numerical Functions Defined on a Topological Space

Theorem 6. If f_1 and f_2 are two positive upper semi-continuous numerical functions, the function h defined by $h(x) = f_1(x)f_2(x)$ is upper semi-continuous.

Proof. If $\alpha > 0$, the set

$$\{x \mid f_1(x)f_2(x) < \alpha\} = \bigcup_{\lambda>0} \{x \mid f_1(x) < \lambda\} \cap \left\{x \mid f_2(x) < \frac{\alpha}{\lambda}\right\}$$

is a union of open sets and so is open. $\square$